

# RYAN THOMAS McCUNE, EIT

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Department of Civil, Construction, and Environmental Engineering  
North Carolina State University · 915 Partners Way, Raleigh, NC 27606

## EDUCATION

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|---------------------------------------------------------------------------------------------------------------|---------------|
| <b>Ph.D. Civil Engineering</b>                                                                                | Expected 2027 |
| North Carolina State University                                                                               | Raleigh, NC   |
| <i>Concentration in Coastal Engineering</i>                                                                   |               |
| <b>Master of Civil Engineering</b>                                                                            | 2025          |
| North Carolina State University                                                                               | Raleigh, NC   |
| <i>Concentration in Coastal Engineering</i>                                                                   |               |
| <b>Honors Bachelor of Environmental Engineering with Distinction</b>                                          | 2022          |
| University of Delaware                                                                                        | Newark, DE    |
| <i>Thesis: Potential Impacts of Soil Aging on TDR Calibrations of Biochar Amended Urban and Coastal Soils</i> |               |
| <b>Honors Bachelor of Civil Engineering</b>                                                                   | 2022          |
| University of Delaware                                                                                        | Newark, DE    |
| <i>Concentration in Facilities Design and Construction</i>                                                    |               |

## PROFESSIONAL EXPERIENCE

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| <b>Graduate Research &amp; Teaching Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2022–present       |
| <i>Department of Civil, Construction, and Environmental Engineering, NC State University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Raleigh, NC        |
| <ul style="list-style-type: none"><li>· Develop high-resolution coastal flood simulations (SFINCS, ADCIRC) and engineer automated Python workflows on HPC clusters to analyze chronic flood hazards.</li><li>· Assemble and deploy custom, low-cost environmental sensor packages (Raspberry Pi, pressure transducers, cameras) and develop rigorous QA/QC data validation frameworks to maintain high-fidelity, publicly accessible flood monitoring datasets.</li><li>· Collaborate on the interdisciplinary Sunny Day Flooding Project, integrating physical hydrodynamic modeling with semantic image segmentation and social impact data to characterize regional risk for coastal communities.</li><li>· Direct technical training for undergraduate researchers in Python programming and machine learning, and deliver guest lectures and instructional support for a Coastal Engineering course of over 80 students.</li></ul> |                    |
| <b>Research Intern</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2022               |
| <i>United States Geological Survey</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | St. Petersburg, FL |
| <ul style="list-style-type: none"><li>· Developed MATLAB scripts to analyze high-frequency optical data from the USGS coastal camera network, extracting and quantifying wave runoff parameters.</li><li>· Applied statistical techniques to transform raw imagery into actionable hydrodynamic metrics, supporting regional coastal change research.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                    |

**Keller Family Senior Writing Fellow***Honors College, University of Delaware*

2021–2022

Newark, DE

- Directed operational logistics and performance reviews for a university fellowship team as the inaugural Keller Family Senior Fellow.
- Managed the professional development of peer educators, providing strategic mentorship and actionable feedback to improve writing pedagogy and communication coaching.

**Engineering Intern***Coastal Resilience Design Studio, University of Delaware*

2020–2022

Newark, DE

- Collaborated within a multidisciplinary team of landscape architects, policy analysts, and engineers to develop community-informed conceptual designs for coastal public infrastructure.
- Integrated technical engineering constraints with aesthetic and regulatory requirements, culminating in a First Place award in the national Coastal & Estuarine Research Federation (CERF) design competition.

**Undergraduate Teaching Assistant***Department of Civil & Environmental Engineering, University of Delaware*

2020–2022

Newark, DE

- Facilitated supplementary instruction and office hours for students, translating theoretical civil and environmental engineering concepts into practical problem-solving strategies.
- Evaluated technical assignments against established rubrics and provided targeted feedback to bridge gaps in student comprehension.

**Undergraduate Research Assistant***Department of Civil & Environmental Engineering, University of Delaware*

2018–2022

Newark, DE

- Quantified the hydraulic performance of biochar-amended soils for stormwater filtration by calibrating and deploying Time Domain Reflectometry (TDR) sensors.
- Conducted independent laboratory analyses to measure the Electron Reduction Capacity (ERC) of biochar substrates for environmental remediation.
- Executed field data collection for transportation planning studies, deploying JAMAR counters to capture and analyze vehicular traffic volumes across several sites.

**Engineering Intern***C.S. Davidson, Inc.*

2020–2021

York, PA

- Analyzed historical state contract datasets to construct updated unit price schedules, improving the accuracy of municipal project cost estimations and bidding.
- Conducted technical reviews of land development plans to ensure strict compliance with municipal ordinances, zoning codes, and stormwater management regulations.
- Executed precision field surveys and stormwater Best Management Practice (BMP) inspections to verify construction quality and document existing infrastructure.

**Writing Fellow***Honors College, University of Delaware*

2020–2021

Newark, DE

- Completed comprehensive training in writing pedagogy and communication strategy to provide advanced editorial guidance across multiple academic disciplines.
- Partnered with university faculty to mentor a cohort of 20 students per semester, reviewing dozens of academic essays to enhance structural argumentation and clarity.

**Munson Fellow***Honors College, University of Delaware*

2019–2020

Newark, DE

- Advised a residential cohort of over 30 first-year Honors students on degree planning, course selection, and navigating university administration to support institutional retention goals.
- Designed, managed logistics for, and executed community-building events and social programming to foster a highly engaged residential learning environment.

**Engineering Intern***Manchester Township*

2019

York, PA

- Digitized legacy engineering archives to establish a centralized inventory of over 300 stormwater Best Management Practices (BMPs) using the CSDatum platform.
- Optimized municipal maintenance operations by structuring infrastructure data to enable efficient tracking of BMP performance, regulatory compliance, and necessary repairs.

**HONORS & AWARDS**

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|-----------------------------------------------------------------------------------------------|-----------|
| National Defense Science & Engineering Graduate Fellowship                                    | 2024–2027 |
| ICCE Student Travel Scholarship                                                               | 2026      |
| KIETS Climate Leaders Program Scholar                                                         | 2024      |
| AGU Outstanding Student Presenter Award                                                       | 2023      |
| Honorable Mention, NSF Graduate Research Fellowship Program                                   | 2023      |
| 2nd Place, NC State EWC Student Poster Competition                                            | 2023      |
| Provost Doctoral Fellowship, North Carolina State University                                  | 2022      |
| Award of Excellence in Student Collaboration, American Society of Landscape Architects (ASLA) | 2022      |
| PA-DE Chapter of the American Society of Landscape Architects Student Honor Award             | 2022      |
| RJN Foundation Department of Civil & Environmental Engineering Award                          | 2022      |
| 1st Place, 2021 Coastal & Estuarine Research Federation Design Competition                    | 2021      |
| Junior Award, Delaware Section of the American Society of Civil Engineers (ASCE)              | 2021      |
| Honors Enrichment Award, University of Delaware Honors College                                | 2021      |
| Chair's Fellowship, University of Delaware Department of Civil & Environmental Engineering    | 2021      |
| General Honors Award, University of Delaware Honors College                                   | 2020      |

**PRESENTATIONS**

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**Oral Presentations**

1. Collins, J., Hino, M., **McCune, R.**, Anarde, K., (2026). *The Price of Paradise: How Communities Tolerate Chronic Coastal Flooding in Rural North Carolina*. Population Association of America Annual Conference. St. Louis, MO, 2026.
2. **McCune, R.**, Anarde, K., Sebastian, A., Collins, J. P., Grimley, L., Hamidi, E., Hino, M., Dietrich, J. C., (Dec. 2025). *The End of the Road: Present and Future Chronic Flood Risk Along Coastal Roadways and Impacts to Community Livability*. Presented at the AGU Fall Meeting. New Orleans, December 2025.

3. **McCune, R.**, Anarde, K., (Aug. 2025). *Evaluation of Chronic Coastal Flooding Inundation of Low-Lying Roadways and Impacts to Community Livability*. Presented at the 3rd Regional (East Coast) Peer Exchange for Sustainable Eco-Resilient Bridges and Structures. Raleigh, August 2025.
4. Collins, J., Hino, M., **McCune, R.**, Anarde, K., Frankenberg, E., (2025). *Tolerating the tide: accommodation and tolerance of chronic coastal flooding in rural North Carolina*. Population Association of America Annual Conference. Washington, D.C., 2025.
5. **McCune, R.**, Anarde, K., Goldstein, E. B., Srebnik, E. R., Thelen, T., Hino, M., (Sept. 2024). *Quantification of chronic coastal flooding using machine learning*. Presented at the International Conference of Coastal Engineering. Rome, September 2024.
6. Hino, M., Anarde, K., **McCune, R.**, Thelen, T., Farquhar, E., Fridell, T., Whipple, T., Woodard, P., (2024). *Incidence and Impacts of Chronic Coastal Flooding in North Carolina*. Invited presentation at the AGU Fall Meeting. Washington, D.C., 2024.
7. **McCune, R.**, Anarde, K., Goldstein, E. B., (Dec. 2023). *Semantic Image Segmentation of Coastal Roadway Inundation*. Presented at the AGU Fall Meeting. San Francisco, December 2023.
8. Muldrow, L., **McCune, R.**, Bruck, J., (Apr. 2022). *Resilient Self-Generative Infrastructure: A Blue Carbon Solution for Coastal Protection in Hampton, VA*. PA-DE ASLA Conference on Landscape Architecture. Wilmington, April 9, 2022.

### **Poster Presentations**

1. **McCune, R.**, Anarde, K., Grimley, L., Collins, J., Hamidi, E., Sebastian, A., Hino, M., Dietrich, C., (Mar. 2026). *Assessing Chronic Flood Risks to Coastal Infrastructure and Communities*. Presented at the NC State Environmental, Water, and Coastal Engineering Symposium. March 6, 2026.
2. **McCune, R.**, Anarde, K., Goldstein, E., Baker, C., (Mar. 2025). *Quantification of Chronic Coastal Flooding: A machine learning-driven approach to water level extraction*. Presented at the NC State Environmental, Water, and Coastal Engineering Symposium. March 21, 2025.
3. **McCune, R.**, Collins, J., Anarde, K., Hino, M., (Sept. 2024). *A Summer Down East: Internship and Research Experiences in Carteret County*. Presented at the KIETS Climate Leaders Symposium 2024. September, 2024.
4. **McCune, R.**, Anarde, K., Hino, M., Frankenburg, E., Amspacher, K., (May 2024). *Rising tides, drowning ditches: Analysis and communication of chronic coastal flooding in rural communities*. Presented at the National Adaptation Forum. Baltimore, May 2024.
5. **McCune, R.**, Anarde, K., Goldstein, E., (Mar. 2024). *Witness to the Rising Tide: Semantic Image Segmentation of Chronic Coastal Flooding*. Presented at the NC State Environmental, Water, and Coastal Engineering Symposium. March 8, 2024.
6. **McCune, R.**, Anarde, K., Goldstein, E., (Mar. 2023). *On-device Machine Learning for Identifying the Spatial Extent of Chronic Coastal Flooding*. Presented at the NC State Environmental, Water, and Coastal Engineering Symposium. March 10, 2023.
7. Anarde, K., Goldstein, E., Bolewitz, J., **McCune, R.**, Gold, A., Hino, M., (Dec. 2022). *On-device machine learning for identifying the spatial extent of chronic coastal floods*. Presented at the International Conference of Coastal Engineering. Sydney, December 5, 2022.
8. **McCune, R.**, Fettke Von Koeckritz, H., Bruck, J., Puleo, J. A., (Oct. 2021). *Fenwick Island Dune Encroachment Monitoring Project*. Presented at the Young Coastal Scientists and Engineers Conference – Americas. Myrtle Beach, October 30, 2021.

### **Other Invited Lectures or Research Presentations (not listed above)**

1. **McCune, R.**, Anarde, K., Hino, M., Nelson, N., (July 2026). *Threats to Coastal Infrastructure from Climate Change*. Presented at the Envisioning Transformations Workshop. July 30, 2026.
2. **McCune, R.**, Anarde, K., Collins, J., Sebastian, A., Hino, M., Dietrich, C., (May 2026). *Below the Waterline: Chronic Flood Risk on Rural Roads and Impacts to Community Livability*. Presented at the 39th International Conference on Coastal Engineering. May 21, 2026.
3. **McCune, R.** (Apr. 2026). *Guest Lecture: Natural and Nature-based Features*. Department of Civil, Construction, and Environmental Engineering. Introduction to Coastal Engineering.
4. **McCune, R.** (Mar. 2025). *Guest Lecture: Natural and Nature-based Features*. Department of Civil, Construction, and Environmental Engineering. Introduction to Coastal Engineering.
5. **McCune, R.** (Mar. 2024). *Guest Lecture: Natural and Nature-based Features*. Department of Civil, Construction, and Environmental Engineering. Introduction to Coastal Engineering.
6. **McCune, R.** (Mar. 2023). *Guest Lecture: Natural and Nature-based Features*. Department of Civil, Construction, and Environmental Engineering. Introduction to Coastal Engineering.

## **PUBLICATIONS**

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### **Peer-Reviewed Journal Articles (Published or Accepted)**

1. Hino, M., Anarde, K., Fridell, T., **McCune, R.**, Thelen, T., Farquhar, E., Woodard, P., Whipple, A., (June 2025). "Land-based sensors reveal high frequency of coastal flooding". en. In: *Communications Earth & Environment* 6.1, pp. 1–6. ISSN: 2662-4435. DOI: 10.1038/s43247-025-02326-w.
2. Naquin, K., Adams, D. R., Bailey, M. M., Brown, L., Diez, M., Kanipe, J., **McCune, R.**, Thelen, T., Hunter, D. L., Cooper, C. B., (Apr. 2025). "Not Empty Rain Gauges: Experienced Hobbyists Fulfilled in a Contributory Project". In: *Citizen Science: Theory and Practice*. DOI: 10.5334/cstp.774.

### **Peer-Reviewed Journal Articles (Under Review & In Preparation)**

1. Collins, J., **McCune, R.**, Hino, M., Anarde, K., Amspacher, K., (in review). "The Price of Paradise: How Rural Communities Tolerate Chronic Coastal Flooding in North Carolina".
2. **McCune, R.**, Anarde, K., Goldstein, E., Baker, C., (in preparation). "Quantification of chronic coastal flooding: a machine-learning driven approach to water level extraction". In preparation for *Water Resources Research*.

### **Datasets**

1. Ku, V., Gabbula, S. R. A. K., **McCune, R.**, Budavi, P., Sigdel, R., Buscombe, D., Favela, J., Shah, M., Goldstein, E. B., Fitzpatrick, S., (2022). "Segmentation Labels for Emergency Response Imagery from Hurricane Barry, Delta, Dorian, Florence, Isaias, Laura, Michael, Sally, Zeta, and Tropical Storm Gordon". In.

## **PROFESSIONAL DEVELOPMENT**

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|----------------------------------------------------------------------------|------|
| Short Course: Hydraulic Structures and Storm Surge Barriers, Galveston, TX | 2026 |
| COPRI Leadership Summit, Reston, VA                                        | 2026 |
| Community Surface Dynamics Modeling System Annual Meeting, Boulder, CO     | 2025 |
| Earth Surface Processes Institute, Boulder, CO                             | 2025 |

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|----------------------------------------------------------------------------|------|
| Short Course: Modeling Compound Flooding with SFINCS, Rome, IT             | 2024 |
| From Ice Sheets to the Coast: Sea-Level Rise Impacts Workshop, Houston, TX | 2024 |
| 5th NOAA AI Workshop, virtual                                              | 2023 |
| Coastal Imaging Research Network Workshop, Duck, NC                        | 2023 |
| Blue Economy Workshop, Morehead City, NC                                   | 2023 |

## PROFESSIONAL SERVICE

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|----------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Chair   EWC Seminar Visiting Student Logistics Committee, NC State University                                                          | 2026      |
| Student Ambassador   NC State Climate and Sustainability Academy                                                                       | 2025      |
| Session Chair and Organizer   "The MacGyver Session" Ocean Sciences Poster Session, AGU Annual Meeting                                 | 2025      |
| Member   EWC Seminar Visiting Student Logistics Committee, NC State University                                                         | 2025      |
| Organizer and Moderator   Panel: Community Responses to Chronic Flooding and Sea-Level Rise Impacts, North Carolina Coastal Conference | 2024      |
| Chair   EWC Seminar Food Committee, NC State University                                                                                | 2024      |
| Member   EWC Seminar Food Committee, NC State University                                                                               | 2023      |
| Student Member   Provost Search Committee, University of Delaware                                                                      | 2022      |
| Engineering Ambassador   College of Engineering, University of Delaware                                                                | 2022      |
| Engineering Ambassador   Dept. of Civil & Environmental Engineering, University of Delaware                                            | 2021–2022 |
| Honors College Ambassador   University of Delaware                                                                                     | 2019–2022 |

## Professional Affiliations

- Member, American Society of Civil Engineers (ASCE)
- Member, American Shore and Beach Preservation Association (ASBPA)
- Member, American Geophysical Union (AGU)
- Member, Tau Beta Pi Engineering Honor Society
- Member, Chi Epsilon Civil Engineering Honor Society

## Manuscript Reviewer

- Geoscientific Model Development (1)

## LEADERSHIP

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|-----------------------------------------------------------------------------------|--------------|
| Vice President   Coasts, Oceans, Ports & Rivers Institute (COPRI) Student Chapter | 2026–present |
| President   Coasts, Oceans, Ports & Rivers Institute (COPRI) Student Chapter      | 2025–2026    |
| Vice President   Coasts, Oceans, Ports & Rivers Institute (COPRI) Student Chapter | 2023–2024    |
| Vice President   Environmental Engineering Student Association                    | 2020–2022    |
| Parliamentarian   Phi Sigma Pi National Honor Fraternity Alpha Eta Chapter        | 2021         |